Expt. No. 6 Date

Page No. 16

Aim :- Write a program to fill an ellipse.

Source Code :-

```
#include <graphics.h>
#include <stdlib.h>
#include <stdio.h>
#include <conio.h>

int main(void)
{
  /* request autodetection */
  int gdriver = DETECT, gmode, errorcode;
  int midx, midy, i;
  int xradius = 100, yradius = 50;

/* initialize graphics & local variables */
  initgraph( &gdriver, &gmode,
             "c:\\tc\\bgi");

/* read result of initialization */
errorcode = graphresult();
  if(errorCode != grOk)  /* an error
                            occured */

printf("Graphics error : %s\n", grapherrormsg
                    (errorcode));
printf(" Press any key to halt ");
getch();
exit(1);
}
```

Teacher's Signature

```
midx = getmaxx()/2;
midy = getmaxy()/2;

/* loops through the fill patterns */

for(i = EMPTY_FILL; i < USER_FILL; i++){
    /* set fill pattern */
    setfillstyle(i, COLORS(12));

    /* draw a filled ellipse */
    fillellipse(midx, midy, xradius,
                yradius);
    getch();
}

/* clean up */
closegraph();
return 0;
}
```

Viva Question :-

Q1 What is Mid Point Ellipse Algorithm?

Ans The midpoint Ellipse Algorithm is a computer graphics algorithm used to draw an ellipse efficiently. It uses decision parameters to select the next pixel & relies on integer calculations to reduce computations.

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Q2 Define Ellipse.

Ans An ellipse is a closed curve in which the sum of the distances from any point on the curve to two fixed points, called foci, is constant. It has a major axis & a minor axis.

Q3 What is Boundary Fill Algorithm?

Ans The boundary Fill Algorithm is a computer graphics technique used to fill a closed area with a specific color. It starts from a seed point & fills neighboring pixels until it reaches the boundary color.

Q4 What is Flood fill / Seed fill Algorithm?

Ans The Flood fill Algorithm fills a connected area with a new color, starting from a selected seed point. It replaces all connected pixels of the target color with the new color until the region is completely filled.

Q5 Difference between Boundary Fill & Flood Fill Algorithm.

Ans Boundary Fill fills an area until it reaches a specified boundary color, while Flood fill replaces all connected pixels of a specified interior color. Boundary Fill uses the boundary color to stop, whereas Flood Fill uses the original fill color to identify the region.

Teacher's Signature

Date

Expt. No. 7. Page No. 19.

Aim:- Write a program to fill a polygon.

Source Code :-

```
#include<graphics.h>
#include<stdlib.h>
#include<stdio.h>
#include<conio.h>

int main(void)
{
  int driver = DETECT, gmode, errorcode;
  int i, x, y;
  int corner[8];
  initgraph(&gdriver, &gmode, "c:\\tc\\bgi");
  errorcode = graphresult();
  if(errorcode != grOK)
    {
      printf("Graphics error :%s\n", grapherrormsg
                                    (errorcode));
      printf("Press any key to halt:");
      getch();
      exit(1);
    }
  x = getmaxx()/4;
  y = getmaxy()/4;

  corner[0] = x;
  corner[1] = y;
```

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```
corner[2] = x+200;
corner[3] = y+0;

corner[4] = x+200;
corner[5] = y+150;
corner[6] = x+0;
corner[7] = y+150;

for(i=EMPTY_FILL ; i<USER_FILL ; i++)
  {
    setfillstyle(i, COLORS(random(15)));
    fillpoly(4, corner);
    getch();
  }
closegraph();
return 0;
}
```

Viva Question

Q1 Explain the illumination model.

Ans The illumination model describes how light interacts with objects to determine their brightness & color in computer graphics. It includes ambient, diffuse, & specular reflection components to create realistic images.

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Q2 What is a spline?

Ans A spline is a smooth curve defined by a set of control points. In computer graphics, splines are used to create smooth shapes, curves & object outlines.

Q3 Define B-Spline curve?

Ans A B-spline (Basis spline) curve is a smooth curve defined by control points & basis functions. It provides local control, meaning changing one control point affects only a part of the curve.

Q4 What are the different ways of specifying spline curve?

Ans Spline curves can be specified using explicit, implicit & parametric representations. In computer graphics, the parametric form is commonly used to represent & draw spline curves.

Q5 What are the important properties of Bezier Curve?

Ans A Bezier curve is defined by control points & always passes through its first & last control points. It lies within the convex full of its control points & provides smooth curves widely used in computer graphics & animation.

Teacher's Signature

Date

Expt. No. 8. Page No. 22.

Aim:- Write a progrem to hoist a flag.

Source Code:-

```
#include <dos.h>
#include <graphics.h>
#include <stdlib.h>
#include <conio.h>
int main(void)
{
  int gdriver = DETECT, gmode, poly[10],
                poly1[10], poly2[10], poly3[10];
  initgraph(&gdriver, &gmode, "C:\\tc\\bgi");
//polygon
  {
    poly[0] = 50;
    poly[1] = 50;
    poly[2] = 60;
    poly[3] = 50;
    poly[4] = 60;
    poly[5] = 400;
    poly[6] = 50;
    poly[7] = 400;
    poly[8] = 50;
    poly[9] = 50;

setfillstyle(SOLID_FILL, WHITE);
fillpoly(4, poly);
}
```

Teacher's Signature

```
// flag
{
    poly1[0] = 60;
    poly1[1] = 50;
    poly1[2] = 320;
    poly1[3] = 50;
    poly1[4] = 320;
    poly1[5] = 100;
    poly1[6] = 60;
    poly1[7] = 100;
    poly1[8] = 60;
    poly1[9] = 50;

setstyle(SOLID_FILL, RED);
fillpoly(4, poly1);
    }
{
poly2[0] = 60;
poly2[1] = 100;
poly2[2] = 320;
poly2[3] = 100;
poly2[4] = 320;
poly2[5] = 150;
poly2[6] = 60;
poly2[7] = 150;
poly2[8] = 60;
poly2[9] = 100;
fillpoly(4, poly2);
}
```

```
{
poly3[0] = 60;
poly3[1] = 50;
poly2[2] = 320;
poly3[3] = 50;
poly3[4] = 320;
poly3[5] = 200;
poly3[6] = 60;
poly3[7] = 200;
poly3[8] = 60;
poly3[9] = 50;

setstyle(SOLID_FILL, RED);
    fillpoly(4, poly3);
    }
{
 setstyle(SOLID_FILL, BLINK);
 pieslice(190, 126, 0, 45, 25);
    pieslice(190, 126, 45, 90, 25);
    pieslice(190, 126, 90, 135, 25);
    pieslice(190, 126, 135, 180, 25);
    pieslice(190, 126, 180, 225, 35);
    pieslice(190, 126, 225, 270, 25);
    pieslice(190, 126, 270, 315, 25);
    pieslice(190, 126, 315, 360, 25);
    }
getch();
closegraph();
return 0;
}
```

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Viva Question

Q1 What are the design issues for multimedia authoring?

A1 The main design issues are selecting suitable media, organizing content, designing user interaction, & maintaining synchronization between audio, video, text & graphics. Compatibility, performance, & ease of use are also important.

Q2 What is panning?

A2 Panning is the technique of moving the audio balance between the left & right speakers. In multimedia, it creates a sense of direction & spatial movement of sound.

Q3 Define mail message?

A3 A mail message is an electronic message sent from one user to another over a network. It can contain text, images, audio, video & attachments.

Q4 What are components of a distributed multimedia system?

A4 The main components are multimedia data source, server, communication networks, clients, & storage systems. They work together to store, transmit & present multimedia content across different devices.

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Date

Expt. No. Page No. 26.

Q3 What are the primary I/O topologies used for multimedia?

Ans The primary I/O topologies include bus, star & ring topologies. They define how devices are connected & how multimedia data is transferred between them.

Teacher's Signature

Date

Expt. No. 9 Page No. 27.

Aim:- Write a progrem to create balloons.

Source Code

```
#include <dos.h>
#include <graphics.h>
#include <stdlib.h>
#include <stdio.h>
#include <conio.h>
int main(void)
{
 int main(void)
  {
   int gdriver = DETECT, gmode, errorcode;
  initgraph(&gdriver, &gmode, "C:\\tc\\bgi");
  while(!kbhit())
   {
  int i,j=0;
  for(i=440; j=0; j>20; j<400; j++, i--).
    {
     int k=180+j;
  setfillstyle(SOLID_FILL, WHITE);
  fillellipse(k,i,50,90);
  fillellipse(k,i+95,5,10);
  line(k+5, i+100, k-20, i+270);
   setfillstyle(SOLID_FILL, RED);
   fillellipse(180, i, 50, 90);
   fillellipse(180, i+95, 5, 10);
   line(185, i+100, 220, i+300);
```

Teacher's Signature

```
setfillstyle(SOLID_FILL, BLUE);
fillellipse(250, i, 50, 90);
fillellipse(250, i+95, 5, 10);
line(250, i+100, 220, i+300);
setfillstyle(SOLID_FILL, GREEN);
fillellipse(310, i+95, 5, 10);
line(310, i+100, 290, i+300);
setfillstyle(SOLID_FILL, MAGENTA);
fillellipse(210, i+60, 50, 90);
fillellipse(210, i+155, 5, 10);
line(210, i+160, 220, i+300);
setfillstyle(SOLID_FILL, YELLOW);
fillellipse(270, i+60, 50, 90);
fillellipse(270, i+155, 5, 10);
line(270, i+160, 220+i+300);

delay(50);
cleardevice();
}
}
getch();
closegraph();
return 0;
}
```

Viva Question

Q: list different color models.

Ans: The different color models used in computer graphics are RGB, CMYK, HSV, HSL & YIQ. RGB mainly used for displays, while CMYK is used for printing.

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Date

Expt. No. Page No. 29

Q2 Difference between database & multimedia database

Ans A database stores structured data such as numbers, text & records. A multimedia database stores & manages multimedia data such as images audio, video, graphics & animations.

Q3 List out the input devices of multimedia.

Ans The input devices used in multimedia include a keyboard, mouse, scanner, microphone, webcam, digital camera, joystick & touchscreen. These devices helps users enters text, images, audio & video into a computer.

Q4 What is the use of control points?

Controls points are used to define & control the shape, direction & smoothness of curves in computer graphics. They helps create & modify curves such as Bezier & B-spline curves.

Q5 Define Projection.

Ans Projection is the process of representing a 3D object on a 2D viewing surface. In computer graphics, the two main types are parallel projection & prespective projection.

Teacher's Signature

Date

Expt. No. 10. Page No. 30.

Aim:- Write a program to implement a walking robot.

Source Code:-

```
#include <graphics.h>
#include <stdlib.h>
#include <stdio.h>
#include <conio.h>
int main(void)
  {
   int gdriver = DETECT, gmode, errorcode;
   initgraph(&gdriver, &gmode, "C:\\tc\\bgi");
   errorcode = graphresult();
   if (errorcode != grOk)
  {
   printf("Graphics error: %s\n", grapherrormsg
                                  (errorcode));
   printf("Press any key to halt:");
   getch();
   exit(1);
  }
int i, x, y, poly[10];
}
 while(!kbhit() && (x = 620))
  {
   for(x=0; x<=620; x=x+10)
    {
     /*f*/
```

Teacher's Signature

```
setfillstyle(SOLID_FILL, COLORS(6));
fillellipse(x+50, 150, 25, 25);
setfillstyle(SOLID_FILL, COLORS(BLACK));
fillellipse(x+65, 145, 3, 3);

/* c */
poly[0] = x+25;
poly[1] = 175;
poly[2] = x+75;
poly[3] = 175;
poly[4] = x+75;
poly[5] = 300;
poly[6] = x+25;
poly[7] = 300;
poly[8] = x+25;
poly[9] = 175

drawpoly(5, poly);
/* b1 */
line(x+50, 200, x+20, 315);
line(x+52, 202, x+20, 320);
/* b2 */
line(x+75, 230, x+90, 280);
line(x+75, 232, x+94, 285);

/* 11 */
line(x+50, 300, x+100, 430);
line(x+57, 300, x+107, 430);
```

```
/*12*/
line(x+46, 300, x+25, 430);
line(x+53, 300, x+32, 430);

/* p1 */
line(x+23, 430, x+55, 430);
line(x+23, 435, x+55, 435);

/*p2*/

line(x+97, 430, x+129, 430);
line(x+97, 435, x+129, 435);
delay(150);
clearadvice();
}
}
}
getch();
closegraph();
return 0;
}
```

Viva Question

Q1 What is JPG?

Ans JPG (Joint Photographic's experts group) is a commonly used image file format that uses lossy compression to reduce file size. It is mainly used for storing digital photographs & colorful images.

Teacher's Signature

Q2 Assuming that a full-color (24-bit per pixel) RGB raster system has a 512 x 512 frame buffer, how many distinct color choices (intensity levels) would be available?

Ans In a 24-bit RGB system, each pixel has 8 bits for each of the three colors RED, GREEN & BLUE.
Therefore, the total distinct colors available are $2^{24} = 16777216$ Colors.

Q3 What is the difference between impact & non-impact printers?

Ans Impact printers print by striking an inked ribbon against paper, for example, dot-matrix printers. Non-impact printers print without striking the paper, for example, inkjet & laser printer.

Q4 What is a projector?

Ans A projector is an output device that displays images, videos & presentations on a large screen or surface. It is commonly used in classrooms, meetings & multimedia presentations.

Date

Expt. No. _____ Page No. 34.

Q5 What are hypermedia documents?

Ans Hypermedia documents are documents that contains different types of media, such as text, images, audio, video, & hyperlinks. They allow users to navigate between related information interactively.

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